A Closed-Form Detection Threshold for Muon Absorption Tomography of Concrete Voids

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Abstract

Muon absorption tomography is increasingly used for non-destructive inspection of concrete infrastructure, but feasibility assessment currently requires case-by-case Monte Carlo simulation (Geant4). No closed-form expression exists for the minimum detectable void size as a function of overburden thickness, detector area, and required statistical significance. Here we derive such a formula from first principles, combining the modified Gaisser sea-level muon spectrum (PDG 2024, normalized to the measured vertical intensity of 70 m^{-2} s^{-1} sr^{-1}), the continuous-slowing-down energy loss parameterization of Groom, Mokhov & Striganov (2001), and the Li & Ma (1983) significance statistic from gamma-ray astronomy. The resulting one-equation tool,

> **$t_{min} = \sigma^2 \times 2R / \Delta R^2$**

gives the minimum exposure time t_{min} for a detector of area A to achieve σ-sigma detection of a void of diameter d embedded in concrete of thickness T and density ρ. We validate the formula against the measured sea-level muon rate (~1 cm^{-2} min^{-1}), published depth-intensity relations, and experimental results from the Thompson et al. (2020) railway tunnel campaign. The analytical prediction represents a statistical floor that is 10–100× more optimistic than practical Geant4-based estimates, a discrepancy we quantify and attribute to spatial reconstruction overhead, detector inefficiency, and systematic backgrounds. We provide practical lookup tables and an open-source Python implementation covering the parameter space relevant to civil engineering inspection (T = 0.5–15 m, d = 2–100 cm). The formula enables rapid go/no-go feasibility assessment without simulation, filling a gap between the experimental muon tomography community and the structural engineering profession.

Keywords: muon tomography, concrete inspection, void detection, non-destructive testing, cosmic ray muons, analytical threshold

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1. Introduction

Cosmic ray muons, produced by the interaction of primary cosmic rays with Earth's atmosphere, arrive at sea level with a flux of approximately 1 cm^{-2} min^{-1} and penetrate meters to kilometers of rock or concrete before being absorbed [1, 2]. This penetrating ability has been exploited for imaging since George's 1955 tunnel overburden measurement [3] and Alvarez's 1970 search for hidden chambers in the Pyramid of Chephren [4]. Recent advances in detector technology have brought muon tomography to the point of practical application in civil infrastructure inspection: railway tunnels [5], bridges [6, 7], dams [8], and nuclear reactor buildings [9].

Two complementary modalities exist. Absorption (transmission) radiography measures the attenuation of the muon flux through the target, detecting density deficits such as voids, cracks, or honeycombing. Scattering tomography, which requires detectors on both sides of the target, measures the Coulomb scattering angle to image high-Z materials such as steel reinforcement [10]. This paper addresses absorption radiography only, which requires a single detector and is the natural modality for void detection.

Despite the growing number of experimental demonstrations, a practical barrier persists: determining whether muon tomography is *feasible* for a given inspection scenario requires running a Monte Carlo simulation — typically Geant4 [11] with the CRY cosmic ray library [12] — specific to the geometry, material, and detector system. Thompson et al. [5] ran Geant4 through a 3D topographic reconstruction of the Alfreton Old Tunnel. Dobrowolska et al. [10] simulated a specific concrete block. The GScan group [7] simulated individual bridge decks. Each assessment starts from scratch.

No general, closed-form expression exists that answers the fundamental engineering question: *Given a concrete structure of thickness T, what is the smallest void I can detect with a detector of area A in an exposure time t?*

In this paper, we derive such a formula from established physics. The derivation combines three well-known results from three different fields:

1. The modified Gaisser parameterization of the sea-level muon energy spectrum (particle physics; PDG [1]),
2. The continuous-slowing-down energy loss in concrete (nuclear physics; Groom, Mokhov & Striganov [13]),
3. The Li & Ma significance statistic for excess counts (gamma-ray astronomy; [14]).

Each component is textbook material in its respective community. Their combination into a single engineering tool has not been published.

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2. Theory

2.1 Sea-Level Muon Spectrum

The differential vertical muon flux at sea level is described by the modified Gaisser formula [1, 15]:

$$dN/(dE\, d\Omega) = 0.14 \times E^{-2.7} \times [\, 1/(1 + 1.1E \cos \theta^*/115) + 0.054/(1 + 1.1E \cos \theta^*/850)\,]$$

in units of $cm^{-2}\ s^{-1}\ sr^{-1}\ GeV^{-1}$, where E is the muon energy in GeV and $\cos\theta^*$ includes the Earth curvature correction [15]:

$$\cos\theta^* = \sqrt{[(\cos^2\theta + 0.0024) / 1.0024]}$$

This formula is valid for $E \gtrsim 1$ GeV. Following Tang et al. [15], we note that below ~$100/\cos\theta$ GeV, pion decay kinematics soften the spectrum relative to the high-energy power-law form. The formula is normalized to reproduce the PDG-quoted vertical muon intensity:

$$I_{vert}(> 1\ GeV) = 70 \pm 7\ m^{-2}\ s^{-1}\ sr^{-1}$$

We apply a global normalization factor of 0.082 to the raw Gaisser formula to match this value. This normalization also reproduces the well-known total sea-level rate of approximately 1 muon $cm^{-2}\ min^{-1}$ when integrated over the upper hemisphere with the $\cos^2\theta$ angular distribution [1].

Limitation: Below ~1 GeV, the Gaisser formula does not account for geomagnetic cutoff effects, which depend on geographic latitude and local magnetic field. For concrete thicknesses $T < 1$ m (where $E_{min} < 0.5$ GeV), the formula becomes unreliable. All quantitative results in this paper for $T < 1$ m should be treated as approximate.

2.2 Energy Loss in Concrete

The mean energy loss rate for muons in matter is [13]:

$$\langle -dE/dx \rangle = a(E) + b(E) \times E$$

where a(E) is the ionization (Bethe-Bloch) contribution and b(E) is the sum of radiative processes (bremsstrahlung, pair production, photonuclear interactions). Both a and b are slowly varying functions of energy. For standard concrete ($\rho = 2.3\ g/cm^3$, $\langle Z \rangle \approx 11$, $Z/A \approx 0.50$), representative values at the Bethe-Bloch minimum are [13]:

$$a = 2.07 \pm 0.05\ MeV/(g/cm^2)$$
$$b = (3.40 \pm 0.17) \times 10^{-6}\ (g/cm^2)^{-1}$$

The muon critical energy, where ionization and radiative losses are equal, is:

$$E_c = a/b \approx 609\ GeV$$

In the continuous-slowing-down approximation (CSDA), the solution of the energy loss equation gives the minimum energy E_{min} for a muon to traverse a slab of density-thickness $X = \rho \times T$:

$$\mathbf{E_{min} = (a/b) \times [\exp(b \times X) - 1]}$$

For $X \ll 1/b$ (i.e., $T \ll E_c/a \approx 300$ m of concrete), this simplifies to $E_{min} \approx a \times X$, recovering the familiar minimum-ionizing approximation.

2.3 Transmitted Muon Rate

The muon count rate through a concrete slab of thickness T, measured by a detector of area A with acceptance half-angle θ_{max}, is obtained by integrating the flux above $E_{min}(T, \theta)$ over the acceptance cone:

$$R(T, A) = A \times \int_0^{\theta_{max}} \int_{E_{min}(T/\cos\theta)}^{\infty} [dN/(dE\, d\Omega)] \times \cos\theta \times 2\pi \sin\theta\, d\theta\, dE$$

The $\cos\theta$ factor accounts for the geometric projection of the detector area. The path length through the slab at zenith angle θ is $T/\cos\theta$, so the energy threshold increases with angle.

2.4 Void Detection Significance

A void of diameter d embedded in a slab of thickness T reduces the effective path length from T to (T − d) along the line of sight through the void. This produces an excess muon count in the void pixel relative to adjacent solid-concrete pixels.

In exposure time t, the expected counts are:

$$N_{on} = R(T - d, A_{pixel}) \times t \quad \text{(void pixel)}$$
$$N_{off} = R(T, A_{pixel}) \times t \quad \text{(reference pixel)}$$

where A_{pixel} is the effective detector pixel area subtending the void.

In the Gaussian limit (large counts), the Li & Ma significance [14] reduces to:

$$\sigma \approx (N_{on} - N_{off}) / \sqrt{N_{on} + N_{off}}$$

For small voids where $\Delta R = R(T-d) - R(T) \ll R(T)$, this simplifies to:

$$\sigma \approx \Delta R \times \sqrt{t} / \sqrt{2 R_{bg}}$$

where $R_{bg} = R(T, A_{pixel})$ is the background rate. Solving for the minimum exposure time:

$$\mathbf{t_{min} = \sigma^2 \times 2 R_{bg} / \Delta R^2}$$

This is the central result. It gives the minimum exposure time as a function of the required significance σ, the background rate R_{bg} (determined by T, A, and ρ), and the rate excess ΔR (determined by the void size d).

2.5 Properties of the Formula

The formula has several notable scaling properties:

- **$t_{min} \propto 1/A$**: Doubling the detector area halves the required time. This is a direct consequence of Poisson counting statistics.
- **$t_{min} \propto \sigma^2$**: Going from 3σ to 5σ increases the time by a factor of $(5/3)^2 \approx 2.8$.
- **$t_{min} \propto 1/\Delta R^2$**: The time scales inversely with the square of the rate contrast. Small voids in thick concrete (low contrast) require disproportionately longer exposures.
- **$t_{min} \propto R_{bg}$**: Higher background rates (thinner slabs) increase the time because the Poisson fluctuations are larger in absolute terms, even though the signal is also larger.

3. Results

3.1 Validation

Open-sky muon rate. Integrating the normalized Gaisser spectrum over the upper hemisphere with $\cos^2\theta$ weighting gives 0.88 muons $cm^{-2}\,min^{-1}$, within 12% of the PDG value of ~1 $cm^{-2}\,min^{-1}$. This confirms the flux normalization.

Depth-intensity relation. The computed transmitted rate through standard concrete follows the expected exponential-like attenuation: R drops from ~540 $\mu/s/m^2$ (no overburden) to ~390 $\mu/s/m^2$ (1 m), ~120 $\mu/s/m^2$ (2 m), ~24 $\mu/s/m^2$ (5 m), and ~7 $\mu/s/m^2$ (10 m). These values are consistent with the depth-intensity relations compiled by Crouch [16] and measured by LVD [17], after converting from meters water equivalent (m.w.e.) to meters of concrete (1 m.w.e. = 43.5 cm of standard concrete at $\rho = 2.3\ g/cm^3$).

Contrast check. For a void occupying ~10% of the path length ($d/T \approx 0.1$), the computed contrast $\Delta R/R_{bg}$ ranges from 13% (T = 1 m) to 21% (T = 10 m). This monotonic increase with depth is physically expected: at greater depths, the muon spectrum is harder (higher average energy), and the fractional energy saving from skipping d meters of concrete is larger relative to the total energy budget.

Thompson et al. (2020) comparison. For a 2 m open shaft in 8 m of rock ($\rho \approx 2.0\ g/cm^3$), our formula predicts t_{min} = 0.3 min (5σ) with a 0.5 m^2 detector. Thompson et al. required approximately 30 min of exposure in practice. The ratio of ~100× is discussed in Section 4.1.

3.2 Practical Lookup Table

Table 1 gives minimum exposure times for selected overburden thicknesses and void diameters (5σ, 1 m^2 detector, $\rho = 2.3\ g/cm^3$). These represent the statistical floor; practical times should be multiplied by the safety factor discussed in Section 4.1.

T \ d	2 cm	5 cm	10 cm	20 cm	50 cm	100 cm
1 m	2 min	15 s	3 s	1 s	<1 s	—
2 m	23 min	4 min	1 min	11 s	1 s	<1 s
3 m	1.7 h	16 min	4 min	1 min	6 s	1 s
5 m	12 h	1.8 h	26 min	6 min	1 min	9 s
10 m	6.3 d	24 h	5.9 h	1.4 h	13 min	3 min
15 m	28 d	4.5 d	1.1 d	6.6 h	1.0 h	14 min

Table 2 gives the minimum detectable void diameter for a 24-hour exposure (5σ, 1 m^2 detector):

Overburden T	d_min (24 hours)
1 m	~1 mm
2 m	3 mm
3 m	5 mm
5 m	1.4 cm
10 m	5.0 cm
15 m	10.6 cm

3.3 Systematic Uncertainties

The dominant sources of uncertainty in the analytical prediction are:

1. **Muon flux normalization** (±10%): The PDG vertical intensity carries a ~10% systematic uncertainty [1]. Since t_min $\propto 1/I_0$, this contributes ±10% to the exposure time.
2. **Concrete density** (±0.1 g/cm^3, i.e., ±4%): Real concrete density varies from 2.2–2.5 g/cm^3 depending on mix design, age, and moisture content. This shifts E_min and therefore the transmitted rate. The effect is largest for thick overburden.
3. **Energy loss parameters** (±5%): The Groom et al. parameters a and b carry uncertainties of ~2.5% and ~5%, respectively. Their combined effect on E_min is ~5%.
4. **Gaisser formula validity** (<1 GeV): Below ~1 GeV, geomagnetic cutoff and atmospheric effects reduce the actual flux below the Gaisser prediction. For T < 1 m (where E_min < 0.5 GeV), results are approximate.

Combining these in quadrature, the total systematic uncertainty on t_min is approximately +30% / −20% for T > 1 m, widening for thinner slabs.

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4. Discussion

4.1 From Statistical Floor to Practical Design: The Safety Factor

The analytical formula gives the *statistical floor* — the minimum time assuming perfect detector efficiency, zero background, perfect knowledge of the geometry, and no spatial reconstruction overhead. In practice, several factors increase the required exposure:

Detector efficiency ($\eta \approx 0.5$–0.9). Real detectors do not register every muon that passes through. Scintillator-based systems achieve $\eta \approx 0.9$; drift chamber systems may have $\eta \approx 0.5$–0.8 depending on gas conditions and trigger thresholds [5, 10]. Effect: $t_{min} \rightarrow t_{min} / \eta$.

Angular resolution and pixelization. The formula assumes a single pixel matched to the void size. In practice, the detector has finite angular resolution, and the void signal is spread across multiple pixels. Spatial reconstruction algorithms (e.g., filtered back-projection, MLEM) incur a statistical penalty that depends on the number of voxels and the regularization scheme. Typical penalties: 5–20× [18].

Background from scattered muons. Low-energy muons that scatter significantly in the concrete may appear in the wrong pixel, creating a diffuse background that reduces contrast. This is not captured in the absorption-only model.

Systematic density variations. Real concrete is not homogeneous. Aggregates, rebar, and moisture gradients create density variations that mimic or mask void signals. The formula assumes a uniform background; real surveys require statistical methods to distinguish voids from natural density variations.

Combined safety factor. Based on comparison with the Thompson et al. [5] result (analytical: 0.3 min; practical: 30 min; ratio ≈ 100) and general experience in counting experiments, we estimate a practical safety factor of:

$k_{safety} \approx 10$–100

such that the practical exposure time is $t_{practical} \approx k_{safety} \times t_{min}$. The lower end ($k \approx 10$) applies to large, high-contrast voids with efficient detectors. The upper end ($k \approx 100$) applies to small voids requiring spatial reconstruction in dense structures.

We recommend using $k_{safety} = 30$ as a default planning value. This gives conservative estimates for go/no-go feasibility decisions.

4.2 Scope and Limitations

This formula applies to *absorption radiography* of voids (density deficits) in concrete. It does not apply to:

- **Scattering tomography**, which uses multiple Coulomb scattering to image high-Z inclusions (e.g., rebar). The physics and statistics are fundamentally different [10].
- **Density anomalies below the noise floor**. The formula assumes a binary void (air vs. concrete). Partial density reductions (e.g., honeycombing, moisture loss) produce smaller contrasts and require proportionally longer exposures.
- **Inclined or irregular geometries**. The formula assumes a planar slab. For curved structures (domes, arches), the path length varies across the detector face, requiring angular binning.
- **Underground or shielded locations**. At sites with significant overburden above the detector (e.g., underground labs), the muon flux is reduced and the spectrum is harder. The

formula can still be applied by replacing the sea-level flux with the appropriate depth-modified spectrum [17, 19].

4.3 Comparison with Existing Approaches

The only existing method for estimating muon tomography feasibility is case-specific Monte Carlo simulation. Table 3 compares the analytical approach with Geant4:

Aspect	Geant4 + CRY	This work
Setup time	Days–weeks	Minutes
Computation time	Hours–days	Seconds
Geometry	Arbitrary 3D	Planar slab
Physics	Full (scattering, straggling, secondaries)	CSDA absorption only
Accuracy	High (detector-specific)	±30% (statistical floor)
Use case	Detailed design	Rapid feasibility screening

The analytical formula does not replace Geant4 for detailed detector design. It serves as a rapid screening tool: if $t_{min} \times k_{safety}$ exceeds the available survey time, the inspection is not feasible and there is no need to run a simulation. If $t_{min} \times k_{safety}$ is well within budget, the inspection is clearly feasible. Geant4 is needed only for marginal cases.

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5. Conclusions

We have derived a closed-form expression for the minimum exposure time required to detect a void of diameter d in concrete of thickness T using muon absorption tomography:

$$t_{min} = \sigma^2 \times 2R(T, A) / [R(T-d, A) - R(T, A)]^2$$

where R(T, A) is the transmitted muon rate computed from the PDG sea-level spectrum and Groom et al. energy loss tables. The formula is built entirely from established physics with no free parameters. It reproduces the known sea-level muon rate to within 12% and is consistent with published experimental results when a practical safety factor of 10–100× is applied.

The key practical results are:

1. A 5 cm void in a 5 m concrete structure requires ~2 hours (statistical floor) with a 1 m^2 detector at 5σ — approximately 3 days with the recommended safety factor of 30.

2. A 10 cm void in a 10 m structure requires ~6 hours (statistical floor), or approximately 1 week practical.
3. Voids smaller than ~1 cm in structures thicker than 5 m require month-scale exposures even with 1 m^2 detectors, placing them at the edge of practical feasibility.

The formula, lookup tables, and open-source Python implementation are provided as supplementary material. We hope this tool will lower the barrier between the muon tomography research community and the civil engineering profession.

Data and Code Availability

The complete Python implementation (muon_void_detection_v3.py), including all figures and validation checks, is available as supplementary material and will be deposited on Zenodo upon publication.

References

[1] Particle Data Group. Review of Particle Physics: Cosmic Rays. Phys. Rev. D 110, 030001 (2024). Section 30.

[2] K. Nagamine. Introductory Muon Science. Cambridge University Press (2003).

[3] E.P. George. Cosmic rays measure overburden of tunnel. Commonwealth Engineer, July 1, 455 (1955).

[4] L.W. Alvarez et al. Search for Hidden Chambers in the Pyramids. Science 167, 832 (1970).

[5] L.J. Thompson et al. Muon tomography for railway tunnel imaging. Phys. Rev. Research 2, 023017 (2020).

[6] GScan. Applications of Muon Flux Tomography in assessment of concrete bridge structures. Proc. NDTCE 2025.

[7] M. Dobrowolska et al. Towards fully autonomous muography of concrete structures. Measurement 177, 109235 (2021).

[8] L. Oláh et al. High-resolution muography of a dam. Sci. Rep. 13, 18364 (2023).

[9] H. Fujii et al. Performance of detecting system for spent nuclear fuel loaded in dry cask using muon scattering image. J. Nucl. Sci. Technol. 57, 401 (2020).

[10] M. Dobrowolska et al. Development of muon scattering tomography for detection of reinforcement in concrete. arXiv:2103.14054 (2021).

[11] S. Agostinelli et al. Geant4—a simulation toolkit. Nucl. Instrum. Methods A 506, 250 (2003).

[12] C. Hagmann, D. Lange, D. Wright. Cosmic-ray shower generator (CRY) for Monte Carlo transport codes. IEEE Nucl. Sci. Symp. Conf. Rec. 2, 1143 (2007).

[13] D.E. Groom, N.V. Mokhov, S.I. Striganov. Muon Stopping Power and Range Tables 10 MeV–100 TeV. At. Data Nucl. Data Tables 78, 183 (2001).

[14] T.-P. Li, Y.-Q. Ma. Analysis methods for results in gamma-ray astronomy. ApJ 272, 317 (1983).

[15] A. Tang, G. Horton-Smith, V.A. Kudryavtsev, A. Sherbrooke. Muon simulations for Super-Kamiokande, KamLAND, and CHOOZ. Phys. Rev. D 74, 053007 (2006).

[16] M. Crouch. An improved world survey expression for cosmic ray vertical intensity vs. depth in standard rock. Proc. 20th ICRC 6, 165 (1987).

[17] LVD Collaboration. Muon "depth-intensity" relation measured by the LVD underground experiment. Phys. Rev. D 58, 092005 (1998).

[18] D. Lussier et al. Muon scattering tomography image reconstruction with MLEM. Nucl. Instrum. Methods A 930, 101 (2019).

[19] L.N. Bogdanova, M.G. Gavrilov, V.N. Kornoukhov, A.S. Starostin. Cosmic muon flux at shallow depths underground. Phys. At. Nucl. 69, 1293 (2006).

[20] E.V. Bugaev et al. Atmospheric muon flux at sea level, underground and underwater. Phys. Rev. D 58, 054001 (1998).

[21] D. Ludwig et al. The muon intensity in the Felsenkeller shallow underground laboratory. Astropart. Phys. 112, 24 (2019).